

CHENG-YOU HO

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EDUCATION

- Yale College:** B.S., Applied Mathematics, Certificate in Data Science Aug 2025 - May 2029
- GPA: 4.00/4.00
 - Intermediate Machine Learning, Trustworthy Deep Learning, Algorithmic Game Theory (Fall 2026), Proof-based Linear Algebra, Introduction to Macroeconomics (Spring 2026), Multivariable Calculus (Fall 2025), .
 - Graph and Geometric Learning Lab, Splash at Yale, Yale Ballroom Dance Team, Yale Poker Club
- National Chiayi Senior High School, Taiwan:** Class of Science Aug 2022 - May 2025
- GPA: 3.96/4.00, SAT: 1550, Valedictorian, Harvard Book Prize, Young Physicists' Tournament Personal Awards

HONORS & AWARDS

- International Physics Olympiad (IPhO) · *Silver Medal* 2025
- Asian Physics Olympiad (APhO) · *Gold Medal* 2025
- MIT iQuHack Quantum Computing Hackathon (Alice and Bob) · *1st Place* 2026
- MOSTRATEC International Science and Technology Fair · *2nd Prize (Representing Taiwan)* 2024
- International Junior Science Olympiad (IJSO) · *Silver Medal* 2022

PROFESSIONAL EXPERIENCE

- Prof. Rex Ying's Lab** *Machine Learning Researcher* New Haven, Aug 2026 - Present
- Working on knowledge graphs memory systems at the Yale Graph and Geometric Learning Lab.
 - Built LLM semantic search pipelines, memory-read mechanisms, and benchmarked retrieval performance.
- Banana Labs, Pre-Seed AI NeoLab** *Co-founder, RL Environments & Evals* New York, Jul 2026 - Aug 2026
- Built containerized agentic RL evaluation tasks in numerical reasoning and math proving for \$4M pre-seed raise.
 - Conducted research on reward signals alongside 6 quantitative co-founders, specializing in high-quality evals.
- Sustentra, Pre-Seed AI Fintech Startup** *Head of Growth* New Haven, Apr 2026 - Aug 2026
- Building an AI enterprise product with a Yale SOM '25 founder focused on corporate carbon compliance.
 - Designed agentic AI GTM pipeline; financial statements for \$2M pre-seed round; website design and tracking.
- Yale Quantum Institute** *Quantum Machine Learning Researcher* New Haven, May 2026 - Present
- Identifying symmetries in quantum ML with LLM-Guided evolutionary search under Professor Steven Girvin.
- Yale Undergraduate Capital Partners (KOMPAS VC)** *VC Analyst* New Haven, Oct 2025 - Present
- Conducted due diligence and startup sourcing in climate tech and evaluated emerging investment opportunities.
 - Contributed to a memo on presented at the 2025 United Nations Climate Change Conference.
- National Taiwan University** *Researcher* Taiwan, Jan 2025 - May 2025
- Analyzed large-scale space-telescope observations (Gaia) using Python to discover new stellar patterns.
 - Performed numerical simulations of a galaxy to quantify the effect of dark matter.

LEADERSHIP & COMMUNITY ENGAGEMENT

- YQuantum** *Co-president* New Haven, Sep 2026 - Present
- Co-president of Yale's premier quantum computing hackathon, in charge of an organizing committee of 20.
- Splash at Yale Fall 2026** *Program Director, Administrator* New Haven, 2025 - Present
- Invited high school and middle school students on campus for classes taught by Yale undergraduates.
 - Co-organized large-scale events hosting 500+ participants; managed logistics and volunteer coordination.

SKILLS & INTERESTS

Skills: Python (PyTorch, Tensorflow), Machine Learning, Software Engineering, AI Engineering, Frontend, SQL, Financial Modeling, Go-To-Market Engineering, Data Science, Mandarin Chinese (Native), English (Fluent)

Interests: Machine Learning, Video Editing, Social Media Management, Graphics Design, Quantum Computing, Climate Tech, Ballroom Dancing, Spikeball, Guitar, Piano, Poker, Photography